Space Visual Dictionary

Five chapters covering

Aerodynamics **Aerospace**
Rockets and Missiles
Drones **Space Science**

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Contents

Aerodynamics

The study of air in motion and the forces it creates on solid bodies. From boundary layers to shock waves, airfoils to wind tunnels.

Aerodynamics The study of how air flows around solid objects and the forces generated by moving air.

Aerodynamic Center The point on an airfoil about which the pitching moment is constant regardless of angle of attack.

Aerodynamic Force The resultant force on a body moving through air, resolved into lift, drag, and side force.

Aerodynamic Heating Frictional heating caused by the compression of air at high speeds.

Aeroelasticity The study of interaction between aerodynamic forces and structural deformation.

Aileron A hinged control surface on the trailing edge of a wing that controls roll.

Airfoil A cross-sectional shape designed to produce lift or control from the air.

Angle of Attack The angle between the chord line of an airfoil and the oncoming airflow.

Angle of Incidence The angle between the wing chord line and the aircraft longitudinal axis.

Anhedral A downward angle of the wings decreasing lateral stability.

Aspect Ratio The ratio of wingspan to mean chord. Higher AR gives lower induced drag.

Attached Flow Airflow that smoothly follows the contour of a surface.

Autorotation The condition where a rotor turns due to airflow passing through it.

Boundary Layer The thin viscous region of flow adjacent to a surface.

Boundary Layer Separation Detachment of the boundary layer from the surface, causing a wake.

Boundary Layer Transition The change from laminar to turbulent flow in the boundary layer.

Camber The curvature of the mean camber line of an airfoil.

Center of Pressure The point on an airfoil where the resultant aerodynamic force acts.

Chord The straight-line distance from the leading edge to the trailing edge of an airfoil.

Chord Line The imaginary line connecting the leading and trailing edges.

Circulation The line integral of velocity around a closed curve, proportional to lift.

Compressible Flow Flow in which density variations are significant.

Critical Angle of Attack The angle of attack at which the wing stalls.

Critical Mach Number The freestream Mach at which sonic flow first appears on the airframe.

Dihedral The upward angle of wings increasing lateral stability.

Downwash The downward deflection of air behind a wing, contributing to induced drag.

Drag The aerodynamic force opposing motion through the air.

Drag Coefficient A dimensionless coefficient relating drag to dynamic pressure and area.

Dynamic Stall A delayed stall from rapid change in angle of attack, with temporary vortex lift.

Dynamic Stability The evolution of aircraft motion over time after a disturbance.

Dutch Roll A coupled lateral-directional oscillation.

Elevator A horizontal control surface on the tail that controls pitch.

Elevon A combined elevator and aileron control surface on tailless aircraft.

Expansion Wave A wave that expands and accelerates supersonic flow around a convex corner.

Flap A high-lift device on the trailing edge that increases camber and lift.

Flow Separation Detachment of boundary layer from a surface.

Form Drag Drag due to the shape of the body.

Friction Drag Drag due to shear stress between the surface and the fluid.

Hypersonic Flow Flow with Mach number greater than 5.

Incompressible Flow Flow in which density is assumed constant.

Induced Drag Drag created as a consequence of producing lift.

Induced Velocity The downward velocity component imparted to the air by a lifting wing.

Interference Drag Drag from the interaction of flow fields between different components.

Inviscid Flow An idealized flow neglecting viscosity effects.

Irrotational Flow Flow in which fluid particles have zero net rotation.

Kutta Condition The principle that flow leaves the sharp trailing edge smoothly.

Kutta-Joukowski Theorem Lift per unit span equals density times velocity times circulation.

Laminar Flow Smooth, orderly fluid motion with distinct parallel layers.

Lateral Stability Stability about the longitudinal axis, including roll.

Leading Edge The forward edge of an airfoil or wing.

Lift The aerodynamic force perpendicular to the relative wind supporting the aircraft.

Lift Coefficient A dimensionless coefficient relating lift to dynamic pressure and area.

Longitudinal Stability Stability about the lateral axis concerning pitch motion.

Mach Angle The angle of the Mach cone: mu = arcsin(1/M).

Mach Cone The cone-shaped wavefront from a supersonic body.

Mean Aerodynamic Chord The chord of an equivalent rectangular wing with identical moment characteristics.

Momentum Thickness A measure of the momentum deficit in the boundary layer.

Navier-Stokes Equations The fundamental equations describing viscous fluid motion.

Neutral Point The point where pitching moment is independent of angle of attack.

Normal Shock A shock wave perpendicular to the flow direction.

Nozzle A duct that accelerates a fluid to produce thrust.

Oblique Shock A shock wave inclined to the direction of supersonic flow.

Parasite Drag All drag not associated with lift production.

Phugoid A long-period longitudinal oscillation exchanging kinetic and potential energy.

Pitching Moment The aerodynamic moment about the lateral axis.

Pitot Tube A tube facing the flow to measure stagnation pressure.

Prandtl-Meyer Expansion Isentropic expansion of supersonic flow around a convex corner.

Pressure Drag Drag from the pressure difference between front and rear of a body.

Propeller A rotating airfoil that converts engine power into thrust.

Reattachment The re-establishment of attached flow after separation.

Relative Wind The direction of airflow relative to a moving airfoil.

Reynolds Number The ratio of inertial to viscous forces in a flow.

Rolling Moment The aerodynamic moment about the longitudinal axis.

Rotor A rotating wing assembly that generates lift and propulsion.

Rudder A vertical control surface on the tail that controls yaw.

Schlieren Photography An optical technique for imaging density gradients in flow.

Separated Flow Flow that has detached from the surface.

Shock Wave A thin region across which flow properties change abruptly.

Short Period Mode A well-damped, high-frequency longitudinal oscillation.

Skin Friction The shear resistance between the surface and the fluid.

Skin Friction Drag Drag due to shear stress on the surface.

Slat A leading-edge high-lift device that extends to increase maximum lift.

Sonic Boom The audible shock wave from a supersonic aircraft reaching the ground.

Spoiler A surface that disrupts lift and increases drag for roll control or speed reduction.

Stall Flow separation from the wing due to excessive angle of attack.

Static Margin The distance between the neutral point and CG, determining static stability.

Static Port An opening used to measure static pressure.

Static Stability The initial tendency to return to equilibrium after a disturbance.

Steady Flow Flow where properties at a point are constant over time.

Subsonic Flow Flow with Mach number less than 1.

Supersonic Flow Flow with Mach number greater than 1.

Sweep Angle The angle between the wing quarter-chord line and the lateral axis.

Thickness Ratio The ratio of maximum thickness to chord of an airfoil.

Thrust The propulsive force moving the aircraft forward.

Thrust Vectoring The ability to redirect engine thrust for attitude control.

Trailing Edge The aft edge of an airfoil or wing.

Transonic Flow Flow in the Mach 0.8-1.2 range with mixed subsonic/supersonic regions.

Trim Drag Drag from deflecting control surfaces to maintain trimmed flight.

Turbofan A ducted jet engine where a fan provides a large portion of thrust.

Turbojet A gas turbine engine producing thrust by accelerating a jet rearward.

Turbulent Flow Chaotic, irregular fluid motion characterized by eddies.

Unsteady Flow Flow where properties at a point vary with time.

Upwash The upward flow ahead of a wing that influences surfaces in front.

Vortex A region of rotating fluid, often forming at wingtips.

Vortex Generator A small vane that energizes the boundary layer to delay separation.

Wake The region of disturbed flow behind a body.

Wall Shear Stress The frictional stress exerted by the flow on the surface.

Washout A wing twist that reduces angle of attack at the tip relative to the root.

Wave Drag Drag from shock wave formation in transonic and supersonic flow.

Wing The primary lifting surface of an aircraft.

Wing Area The planform surface area of the wing used in lift/drag calculations.

Wing Span The distance from one wingtip to the other.

Winglet A vertical tip device that reduces induced drag by modifying the tip vortex.

Wing Root The junction where the wing meets the fuselage.

Wing Tip The extreme outer end of the wing.

Yawing Moment The aerodynamic moment about the vertical axis.

Adverse Yaw The tendency to yaw opposite to the roll direction due to differential drag.

Airbrake A device that increases drag to decelerate an aircraft in flight.

Boundary Layer Control Methods to manipulate the state of the boundary layer.

Choked Flow A duct condition where mass flow is limited by the throat reaching Mach 1.

Computational Fluid Dynamics Numerical methods to solve fluid flow problems.

Contra-Rotating Propeller Two coaxial propellers rotating in opposite directions.

Diffuser A duct that decelerates flow, increasing its pressure.

Displacement Thickness The outward displacement of external flow by boundary layer slowing.

Divergence An instability where a disturbance increases in magnitude.

Euler Equations The Navier-Stokes equations without viscosity and thermal terms.

Finite Element Method A numerical technique for approximate boundary value solutions.

Finite Volume Method A numerical method using control volumes to solve PDEs.

Force Balance A device measuring aerodynamic forces on a wind tunnel model.

Grid The discretized representation of a fluid domain.

LES Large Eddy Simulation, solving for large-scale turbulent eddies.

Mach Number The ratio of flow speed to the local speed of sound.

Mesh The cells or nodes forming a discretized fluid volume.

Particle Image Velocimetry An optical flow measurement technique using particle imaging.

Ramjet An air-breathing engine using forward motion for compression.

RANS Reynolds-Averaged Navier-Stokes, a time-averaged CFD approach.

DNS Direct Numerical Simulation, solving all turbulent scales directly.

Scramjet A supersonic ramjet with supersonic internal flow.

Smoke Visualization Using smoke to visualize flow patterns over a model.

Static Test Structural testing under predetermined loads.

Test Section The wind tunnel area where models are placed in uniform flow.

Transitional Flow Flow between laminar and turbulent states.

Turbulence Model A model for Reynolds stresses in turbulent CFD.

Vortex Shedding Alternating vortices shed from a bluff body in a flow.

Wind Tunnel An experimental facility generating controlled airflow.

Aerothermal Heating Heating of a body due to high-speed atmospheric passage.

Ablation Surface material removal by vaporization for thermal protection.

Re-entry The return of a spacecraft to the atmosphere.

Shock Layer The hot shocked gas between the bow shock and vehicle surface.

Thermal Protection System Materials protecting a spacecraft from re-entry heating.

Pulsed Detonation Engine An engine using detonation waves for propulsion.

Turbine A rotary device extracting energy from flowing fluid.

Compressor A device increasing the pressure of a gas.

Spoileron A spoiler used differentially for roll control.

Strake A narrow extension to a wing or fuselage for aerodynamic effect.

Canard A small forewing ahead of the main wing for control or stability.

Blended Wing Body An aircraft configuration blending wing and fuselage.

Flying Wing A tailless aircraft where the wing is the primary structure.

Delta Wing A triangular planform wing.

Swept Wing A wing angled rearward from the root.

Forward Swept Wing A wing angled forward from the root.

Ogive A pointed, curved shape used for nose cones.

Streamline A line tangent to the velocity vector at every point in a flow.

Pathline The trajectory traced by a fluid particle over time.

Streakline The locus of particles passing through a fixed point.

Continuity Equation Mass conservation in fluid flow.

Momentum Equation Newton's second law applied to fluid motion, stating that the net force acting on a fluid element equals the rate of change of its momentum. In integral form, it relates the sum of surface and body forces to the time rate of change of momentum within a control volume plus the net flux of momentum across its boundaries. The momentum equation is fundamental in aerodynamic force analysis and propulsion system design.

Energy Equation Energy conservation in fluid flow.

Bernoulli Equation Constant total pressure along a streamline in inviscid flow.

Camber Line The line midway between the upper and lower airfoil surfaces.

Mean Line Another term for the camber line of an airfoil.

Pressure Coefficient A dimensionless pressure normalized by dynamic pressure.

Stagnation Point A point on a body where the flow velocity is zero.

Stagnation Pressure The pressure at a stagnation point.

Static Pressure The ambient pressure of the flow.

Total Pressure The sum of static and dynamic pressure.

Aerodynamic Twist Variation of airfoil section angle along the span.

Geometric Twist Physical twist of the wing structure.

Wash-In A wing twist increasing angle of attack at the tip.

Oswald Span Efficiency A factor relating induced drag to that of an elliptical wing.

Elliptical Wing A wing with elliptical planform, giving minimum induced drag.

Taper Ratio The ratio of tip chord to root chord.

Wing Loading The ratio of aircraft weight to wing area.

Load Factor The ratio of lift to weight, measured in g.

V-n Diagram A plot of airspeed versus load factor defining flight envelope.

Maneuverability The ability of an aircraft to change direction quickly.

Agility A measure of rapid attitude change capability.

Turning Radius The radius of a turn at a given speed and bank angle.

Rate of Turn The angular velocity of an aircraft in a turn.

Coordinated Turn A turn with no sideslip, balanced lift and centripetal force.

Sideslip The angle between an aircraft's longitudinal axis and the relative wind direction, indicating a lateral component of motion relative to the airflow. Sideslip affects the distribution of aerodynamic forces on the fuselage and vertical tail, and is a critical parameter in lateral-directional stability analysis, crosswind landings, and coordinated turn calculations.

Ground Effect Reduced induced drag when flying close to the ground.

Reynolds Analogy Relation between heat transfer and skin friction.

Prandtl Number The ratio of momentum diffusivity to thermal diffusivity.

Mach Tuck A nose-down pitching moment at transonic speeds.

Shock Stall Stall caused by shock-induced boundary layer separation.

Buffeting Vibration from unsteady flow separation.

Flutter An aeroelastic oscillation that can be destructive.

Control Reversal Loss or reversal of control effectiveness at high speeds.

Gust Load A transient load from atmospheric turbulence.

Maneuvering Load Load from pilot-induced maneuvers.

Lift Distribution Spanwise variation of lift on a wing.

Trailing Vortex The vortex rolling up from the wingtip.

Starting Vortex The vortex shed when a wing begins moving.

Horseshoe Vortex The vortex system modeled as bound vortex plus trailing vortices.

Panel Method A numerical potential flow method using surface panels.

Vortex Lattice Method A numerical method modeling surfaces with vortex rings.

Source Panel Method A panel method using source/sink distributions.

Doublet A potential flow element combining a source and sink.

Rankine Body A streamlined body shape from source-sink superposition.

Joukowski Airfoil An airfoil shape generated by conformal mapping.

Thin Airfoil Theory Linearized theory predicting airfoil lift and moment.

Lifting Line Theory Prandtl's classical aerodynamic theory for finite wings that models the wing as a bound vortex line with trailing vortices forming a vortex sheet. The theory predicts spanwise lift distribution, induced drag, and downwash, and shows that an elliptical lift distribution produces minimum induced drag.

It provides the theoretical basis for understanding finite wing aerodynamics and wing design.

Lifting Surface Theory A more general theory for wings of arbitrary planform.

Sweep Theory Simplified method for swept wing aerodynamics.

Slender Body Theory Theory for long, thin bodies at small angles of attack.

Area Rule A transonic aerodynamic design principle developed by Richard Whitcomb stating that the cross-sectional area of an aircraft should vary smoothly along its length to minimize wave drag at transonic speeds. The area rule led to the characteristic 'Coke bottle' fuselage shape of transonic aircraft and is critical for optimizing aerodynamic performance near Mach 1.

Supercritical Airfoil An airfoil designed to delay drag rise at transonic speeds.

Shock-Free Airfoil An airfoil designed to avoid shock waves at design conditions.

Natural Laminar Flow A design approach to maintain laminar boundary layer over more of the wing.

Laminar Flow Control Active methods to maintain laminar boundary layer.

Riblet Microgrooves on a surface that reduce skin friction drag.

Dimple A surface indentation used for drag reduction, as on golf balls.

Base Drag Drag from the low-pressure area at a blunt base.

Boattail A tapered aft body contour to reduce base drag.

Bump A protuberance on a surface for flow control.

Fence A vertical surface on a wing to control spanwise flow.

Wing Fence A fence on the upper wing surface to prevent spanwise flow.

Dogtooth A notch in the leading edge to generate vortices.

Leading Edge Extension A strake extending forward from the wing root.

LEX Fence A fence on the leading edge extension for vortex control.

Strakes A sharp-edged surface for vortex generation.

Chine A sharp longitudinal edge on a fuselage for lift.

Belly Flap A flap on the underside for drag or lift control.

Deceleron A split aileron that also acts as an airbrake.

Flapperon A combined flap and aileron.

Taileron A stabilator used differentially for roll control.

Stabilator A one-piece horizontal tail that pivots for pitch control.

V-Tail A tail configuration with two angled surfaces combining rudder and elevator.

Cruciform Tail A tail with horizontal and vertical surfaces forming a cross.

T-Tail A tail with the horizontal surface at the top of the vertical fin.

H-Tail A twin vertical tail configuration.

Twin Tail Two separate vertical fins.

Endplate A flat plate at the wingtip to reduce induced drag.

Hoerner Tip A rounded, drooped wingtip design for drag reduction.

Raked Wingtip A wingtip with a swept trailing edge.

Tip Fence A small vertical surface at the wingtip.

Wingtip Vortex The spiral vortex shed from a wingtip.

Contrail A condensation trail from aircraft engine exhaust.

Vapor Cone A visible cone of condensed water vapor around an aircraft at transonic speeds.

Shock Diamond Visible diamond-shaped patterns in supersonic exhaust plumes.

Mach Wave A weak wave at the Mach angle in supersonic flow.

Bow Shock A detached shock wave ahead of a blunt body.

Attached Shock A shock attached to a sharp leading edge.

Lambda Shock A lambda-shaped shock in transonic flow over airfoils.

Fish-Tail Shock A shock pattern resembling a fish tail.

Detached Shock A shock wave not in contact with the body.

Expansion Fan A fan of expansion waves around a convex corner.

Prandtl-Meyer Function The angle function for Prandtl-Meyer expansion.

Method of Characteristics A mathematical method for solving supersonic flow.

Supersonic Wind Tunnel A wind tunnel designed for Mach numbers above one.

Hypersonic Wind Tunnel A wind tunnel for Mach numbers above five.

Shock Tunnel A tunnel using shock waves to produce high-enthalpy flow.

Arc-Heated Wind Tunnel A tunnel using an electric arc to heat the test gas.

Cryogenic Wind Tunnel A tunnel using cold nitrogen gas to increase Reynolds number.

Indraft Wind Tunnel A tunnel drawing air from the room through the test section.

Open Circuit Tunnel A wind tunnel with no return circuit.

Closed Circuit Tunnel A wind tunnel with a continuous return circuit.

Low Speed Tunnel A wind tunnel for Mach numbers below 0.3.

High Speed Tunnel A wind tunnel for transonic and supersonic testing.

Wall Interference The effect of wind tunnel walls on measured results.

Blockage The ratio of model frontal area to tunnel cross-section.

Turbulence Intensity A measure of the fluctuation level in wind tunnel flow.

Flow Angularity The deviation of flow direction from the tunnel axis.

Sting Mount A support strut holding a model from behind.

Strut Mount A support structure holding a model from below or above.

Magnetic Suspension A model support system using magnetic fields.

Pressure Sensitive Paint A paint that fluoresces in proportion to surface pressure.

Temperature Sensitive Paint A paint indicating surface temperature.

Infrared Thermography An imaging technique for surface temperature measurement.

Optical Strain Measurement A non-contact technique using cameras to measure deformation.

Accelerometer A sensor measuring acceleration for vibration testing.

Strain Gauge A sensor measuring strain on a surface.

Thermocouple A temperature sensor used in aerodynamic testing.

Balance A multi-component force measurement device.

External Balance A balance mounted outside the wind tunnel.

Internal Balance A balance mounted inside the model.

Half Model Testing Testing a half-span model against a wall.

Full Model Testing Testing a complete model.

Dynamic Testing Testing with time-varying conditions.

Static Testing Testing under steady conditions.

Oscillatory Testing Testing with oscillating model or flow.

Grid Turbulence Turbulence generated by a grid in the tunnel.

Hot Wire Anemometry A measurement technique using heated wires for velocity.

Laser Doppler Velocimetry An optical velocity measurement technique.

Phase Doppler Anemometry A technique measuring particle size and velocity.

Pressure Scanner A multi-port pressure measurement system.

Wind Tunnel Fan The fan driving airflow in a wind tunnel.

Honeycomb Straightener A flow conditioning device consisting of a bundle of small-diameter tubes or cells placed upstream of the contraction cone in a wind tunnel. It reduces turbulence and swirl by breaking up large-scale eddies and aligning the flow with the tunnel axis, improving flow quality in the test section.

Settling Chamber The wide-section region of a wind tunnel located between the flow conditioning screens and the contraction cone. It provides a low-velocity environment where turbulence decays and flow uniformity is established before acceleration into the test section.

Contraction Cone The converging section of a wind tunnel that accelerates the flow from the settling chamber to the test section. The contraction ratio (area ratio from inlet to outlet) determines the velocity increase and flow uniformity, with higher ratios producing lower turbulence levels.

Diffuser Section The expanding section of a wind tunnel located downstream of the test section, designed to decelerate the flow and recover static pressure. A well-designed diffuser minimizes pressure losses and improves overall tunnel efficiency.

Corner Vanes Curved guide vanes installed in the corners of closed-circuit wind tunnels to turn the flow with minimal pressure loss and flow separation. They are essential for maintaining flow quality and reducing the power required to drive the tunnel.

Turning Vanes Aerodynamic surfaces placed in duct bends or wind tunnel corners to redirect airflow efficiently. They prevent flow separation and reduce total pressure losses by guiding the flow around turns.

Screens Fine-mesh grids placed in the settling chamber of a wind tunnel to reduce turbulence intensity and improve flow uniformity. Multiple screens with decreasing mesh size are often used in series to achieve high-quality flow.

Acoustic Testing A specialized wind tunnel testing methodology focused on measuring noise generated by aerodynamic flows. It requires anechoic or treated test sections with low background noise levels to accurately capture airframe, engine, and propeller noise signatures.

Aeroacoustics The branch of acoustics that studies noise generation by turbulent fluid motion or aerodynamic forces interacting with surfaces. It encompasses jet noise, airframe noise, propeller and rotor noise, and fan noise, and is critical for aircraft certification and community noise reduction.

Jet Noise The noise generated by the high-velocity exhaust of a jet engine, resulting from turbulent mixing of the jet plume with the ambient air. Jet noise is a major component of aircraft noise during takeoff and is reduced through chevron nozzles and serrated ejectors.

Airframe Noise The aerodynamic noise produced by the airframe of an aircraft during flight, including noise from landing gear, flaps, slats, and wing trailing edges. As engines become quieter, airframe noise has become a dominant contributor to overall aircraft noise during approach and landing.

Propeller Noise The noise generated by rotating propellers due to blade loading, thickness effects, and tip vortices. It consists of tonal components at the blade passage frequency and broadband noise from turbulent boundary layer interactions.

Rotor Noise The acoustic signature of helicopter main rotors and tail rotors, including blade-vortex interaction noise, high-speed impulsive noise, and broadband turbulence ingestion noise. Rotor noise is a critical consideration for urban air mobility and military operations.

Fan Noise The noise produced by the fan stage of turbofan engines, comprising tonal noise at the blade passage frequency and broadband noise from turbulent interactions. Fan noise is a significant contributor to aircraft noise during takeoff and approach.

Combustion Noise The noise generated by unsteady heat release and turbulent mixing in the combustion chamber of a gas turbine engine. Combustion noise contributes to the overall engine noise signature and is influenced by fuel-air mixing and combustor geometry.

Noise Certification The regulatory process by which aircraft are certified to meet maximum allowable noise levels established by aviation authorities such as ICAO, FAA, and EASA. Noise certification defines measurement procedures, flight test conditions, and noise limits for takeoff, approach, and sideline.

Noise Reduction Techniques and technologies employed to reduce aircraft noise, including engine nacelle treatments, chevron nozzles, acoustic liners, shielding by airframe structures, and operational procedures such as continuous descent approaches.

Chevron Nozzle A jet engine exhaust nozzle with a serrated or sawtooth trailing edge that promotes mixing between the hot exhaust jet and the ambient air. Chevrons reduce jet noise by breaking up large-scale turbulent structures and shifting the noise spectrum to higher frequencies.

Hush Kit A set of modifications applied to an aircraft engine or nacelle to reduce noise emissions, typically consisting of acoustic liners, exhaust mixers, and nozzle treatments. Hush kits allow older aircraft to meet modern noise certification standards.

Emission Index A quantitative measure of pollutant mass produced per unit mass of fuel burned in an aircraft engine, expressed in grams of pollutant per kilogram of fuel. Emission indices for NOx, CO, unburned hydrocarbons, and soot are used to assess engine environmental performance.

Carbon Footprint The total amount of carbon dioxide (CO2) emissions attributable to an aircraft or aviation operation over a given period or mission. Aviation carbon footprint reduction strategies include sustainable aviation fuels, improved aerodynamics, lightweight structures, and optimized flight operations.

Sustainable Aviation Fuel A liquid fuel derived from renewable or waste feedstocks such as used cooking oil, agricultural residues, or municipal waste, designed to be a drop-in replacement for conventional jet fuel. Sustainable aviation fuel can reduce lifecycle CO2 emissions by up to 80% compared to fossil-based jet fuel.

Hydrogen Propulsion The use of hydrogen as a fuel for aircraft propulsion, either burned in modified gas turbine engines or used in fuel cells to power electric motors. Hydrogen offers zero CO2 emissions at the point of use but presents challenges in storage volume, cryogenic handling, and infrastructure.

Electric Propulsion Aircraft propulsion systems that use electric motors powered by batteries, fuel cells, or hybrid generators to drive propellers or fans. Electric propulsion enables distributed propulsion configurations, reduces local emissions, and opens new vehicle architectures for urban air mobility.

Hybrid Propulsion A propulsion architecture combining a conventional thermal engine (turbine or piston) with an electric motor and battery system. Hybrid propulsion allows the thermal engine to operate at optimum efficiency while the electric motor provides peak power for takeoff and climb.

Solar Powered Flight Aircraft propulsion using photovoltaic cells mounted on the wing and fuselage surfaces to convert sunlight into electrical energy for motors and batteries. Solar powered aircraft can achieve sustained flight at high altitudes for extended durations, limited by energy storage and diurnal cycles.

High Altitude Flight Flight operations at altitudes above 50,000 feet, where low air density requires high aerodynamic efficiency and specialized propulsion systems. High altitude flight includes stratospheric platforms for communications, surveillance, and atmospheric science.

Stratospheric Flight Flight within the stratosphere (approximately 35,000 to 160,000 feet), characterized by stable atmospheric conditions, low turbulence, and minimal weather. Stratospheric platforms enable long-endurance missions for earth observation, telecommunications, and scientific research.

Suborbital Flight A trajectory that reaches space (above the Karman line at 100 km altitude) but lacks the velocity to achieve orbit, resulting in a ballistic return to Earth. Suborbital flight is used for sounding rockets, crewed space tourism, and hypersonic research.

Atmospheric Re-entry The phase of a spacecraft's return trajectory from space through a planetary atmosphere, characterized by hypersonic velocities, intense aerodynamic heating, and high deceleration loads. Atmospheric re-entry requires thermal protection systems and careful trajectory design to ensure survival.

Re-entry Corridor The allowable range of flight path angles within which a re-entering spacecraft must fly to achieve a safe landing at the intended site. Deviations outside the corridor result in excessive heating, structural failure, or landing far from the target.

Re-entry Angle The angle between the flight path of a re-entering vehicle and the local horizontal at the point of atmospheric entry. A steep re-entry angle increases peak heating rates and deceleration, while a shallow angle extends the heating duration and may cause skip-out.

Skip Re-entry A re-entry trajectory in which the vehicle descends into the upper atmosphere, generates aerodynamic lift to ascend back out of the atmosphere, then re-enters a second time. Skip re-entry extends the range and allows higher entry velocities while managing peak heating.

Ballistic Re-entry A non-lifting re-entry trajectory in which the vehicle follows a purely gravitational path with no aerodynamic lift. Ballistic re-entry produces high peak deceleration and heating rates but is simpler and was used by early crewed capsules such as Mercury and Apollo.

Lifting Re-entry A re-entry trajectory that uses aerodynamic lift generated by the vehicle body or wings to control the descent path and reduce peak deceleration. Lifting re-entry, used by the Space Shuttle and proposed for crewed Mars missions, enables cross-range maneuvering and gentler deceleration.

Re-entry Velocity The speed at which a spacecraft enters a planetary atmosphere, typically between 7.8 km/s (low Earth orbit) and 11.2 km/s (escape velocity from Earth). The re-entry velocity determines the kinetic energy that must be dissipated as heat during atmospheric passage.

Re-entry Heating Rate The rate at which thermal energy is transferred to a spacecraft surface during atmospheric re-entry, typically expressed in watts per square centimeter (W/cm2). The peak heating rate occurs at the point of maximum deceleration and drives thermal protection system design.

Peak Heating The maximum instantaneous heating rate experienced by a spacecraft during atmospheric re-entry, occurring when the combination of velocity and atmospheric density produces the highest convective and radiative heat flux. Peak heating determines the required thickness and material of the thermal protection system.

Heat Flux The rate of thermal energy transfer per unit area, expressed in watts per square meter (W/m2), at a surface exposed to aerodynamic heating. Heat flux is a critical parameter in the design of thermal protection systems and is composed of convective, radiative, and catalytic components.

Convective Heating The component of aerodynamic heating resulting from the transfer of thermal energy from the hot boundary layer gas to the vehicle surface through conduction and convection. Convective heating dominates at moderate re-entry velocities and is influenced by boundary layer state (laminar or turbulent).

Radiative Heating The component of aerodynamic heating resulting from thermal radiation emitted by the high-temperature shock layer gas, particularly at hypersonic velocities above 10 km/s. Radiative heating becomes dominant at very high re-entry velocities and is a major consideration for Mars return and planetary entry.

Catalytic Heating The additional heating that occurs when atomic species (oxygen and nitrogen) recombine on a spacecraft surface during re-entry, releasing chemical energy. Catalytic heating depends on the surface material's recombination efficiency and can significantly increase total heat flux.

Thermal Protection Material Materials specifically engineered to withstand the extreme heat loads of atmospheric re-entry or hypersonic flight. Thermal protection materials include ablative composites (carbon-phenolic), reusable ceramics (tiles), and advanced carbon-carbon composites, each selected for specific temperature ranges and mission profiles.

Ablative TPS A thermal protection system that dissipates heat through the controlled ablation (surface removal) of a sacrificial material, carrying heat away from the underlying structure. Ablative TPS is used for high-heat-flux applications such as ICBM re-entry vehicles and planetary probes.

Reusable TPS A thermal protection system designed to withstand multiple re-entry cycles without replacement, typically using ceramic tiles, fibrous insulation blankets, or metallic panels. Reusable TPS was developed for the Space Shuttle and is essential for affordable reusable launch vehicles.

Insulation Materials with low thermal conductivity placed between the hot outer surface and the underlying structure to reduce heat transfer. In aerospace applications, fibrous insulation blankets, aerogels, and multilayer insulation are used to protect sensitive components from high temperatures.

Thermal Barrier Coating A thin layer of ceramic material applied to metallic components to reduce heat transfer and protect against oxidation and thermal fatigue. Thermal barrier coatings are used on turbine blades, combustion chambers, and exhaust nozzles in gas turbine engines.

Hot Structure An aerospace structural design philosophy in which the primary load-bearing structure is allowed to reach high temperatures during flight, eliminating the need for heavy insulation. Hot structures are typically made of high-temperature alloys, titanium, or ceramic matrix composites.

Cold Structure An aerospace structural design approach in which the primary load-bearing structure is protected from high temperatures by external insulation, maintaining it at relatively low temperatures. Cold structures allow the use of conventional materials like aluminum but add insulation weight.

Nose Cap The forward-most component of a re-entry vehicle or hypersonic aircraft, exposed to the highest stagnation temperatures and pressures. The nose cap is typically made of reinforced carbon-carbon or an ablative material and must withstand extreme thermal and mechanical loads.

Leading Edge TPS The thermal protection system applied to the leading edges of wings, fins, and control surfaces of re-entry vehicles and hypersonic aircraft. Leading edge TPS must withstand high heat fluxes while maintaining aerodynamic shape and structural integrity.

Tiles Rigid ceramic insulating blocks used as reusable thermal protection on the Space Shuttle and other re-entry vehicles. Tiles are made of high-purity silica fibers and are coated with a borosilicate glass coating for waterproofing and emissivity control.

Reinforced Carbon Carbon A composite material consisting of carbon fibers embedded in a carbon matrix, capable of withstanding temperatures up to 3000F (1650C). Reinforced carbon-carbon is used for the highest-temperature areas of re-entry vehicles, including nose caps and wing leading edges.

Flexible TPS A lightweight, conformable thermal protection material made of ceramic or polymer fibers, used for areas with moderate heating on re-entry vehicles. Flexible TPS blankets can be draped over curved surfaces and are easier to install than rigid tiles.

Inflatable TPS A deployable thermal protection system consisting of a flexible, inflatable aeroshell that decelerates a payload during atmospheric entry. Inflatable TPS enables larger drag areas than rigid aeroshells of the same mass, reducing peak heating and deceleration.

Aeroshell The protective outer shell of a spacecraft that provides aerodynamic deceleration and thermal protection during atmospheric entry. An aeroshell typically consists of a heat shield on the forward face and a backshell covering the aft side, enclosing the payload.

Heat Shield The forward-facing component of an aeroshell that absorbs and dissipates the intense heat generated during atmospheric entry. Heat shields may be ablative (consumed during entry) or reusable (ceramic tiles) depending on the mission requirements.

Buoyancy The upward force exerted by a fluid on a body immersed in it, equal to the weight of the fluid displaced by the body (Archimedes' principle). In aerodynamics, buoyancy is the principle behind lighter-than-air vehicles such as balloons and airships, and is proportional to the density difference between the lifting gas and the surrounding air.

Clean Configuration The flight configuration of an aircraft in which all high-lift devices (flaps, slats) and landing gear are fully retracted, minimizing drag and maximizing airspeed for a given power setting. Clean configuration is used for cruise, climb, and high-speed flight.

Decalage The angular difference between the chord lines of the upper and lower wings of a biplane, measured in degrees. Positive decalage (upper wing at a higher angle of incidence) results in greater lift from the upper wing and is used to tailor stall characteristics and longitudinal stability.

Disk Loading The average pressure change across an actuator disk such as a propeller or rotor, defined as the thrust divided by the disk area. Low disk loading is characteristic of rotors and high-lift propellers, while high disk loading is typical of high-speed propellers and ducted fans.

Enstrophy A quantity in fluid dynamics that measures the integrated square of the vorticity magnitude, representing the dissipation of kinetic energy in turbulent flows. Enstrophy is particularly useful in the study of turbulence and is directly related to the turbulent kinetic energy dissipation rate.

Keel Effect The aerodynamic phenomenon where lateral area located behind the center of gravity provides directional stability, analogous to the keel of a ship. In aircraft design, the keel effect is provided by the vertical tail and aft fuselage and contributes to weathercock stability.

Rogallo Wing A flexible, delta-shaped wing design consisting of two conically shaped fabric surfaces held taut by a rigid frame or inflated tubes. Rogallo wings are used in hang gliders, parawings, and some spacecraft recovery systems for their light weight and deployability.

Viscosity A measure of a fluid's resistance to shear deformation, arising from intermolecular forces and momentum transfer between fluid layers. In aerodynamics, viscosity governs the behavior of the boundary layer, determines skin friction drag, and influences flow separation, transition to turbulence, and heat transfer.

Aerospace

Aircraft and spacecraft systems: avionics, propulsion (non-rocket), flight controls, airframe structures, and the engineered systems that enable flight.

Aerospace Engineering The primary field covering aircraft and spacecraft development.

Aircraft Any vehicle designed to travel through the air.

Airplane A powered fixed-wing aircraft.

Helicopter A rotary-wing aircraft with one or more horizontal rotors.

Rotorcraft A class of heavier-than-air craft using rotating blades.

Tiltrotor An aircraft converting between helicopter and airplane modes.

Gyrocopter A rotorcraft with an unpowered rotor in autorotation.

Glider A fixed-wing aircraft without an engine.

UAV An unmanned aerial vehicle flown without onboard pilot.

UAS An unmanned aircraft system including ground station and data link.

eVTOL An electric vertical takeoff and landing aircraft.

Airship A powered, steerable lighter-than-air craft.

Balloon An unpowered lighter-than-air craft.

Spacecraft A vehicle operating beyond Earth's atmosphere.

Satellite An artificial object in orbit around a celestial body.

Launch Vehicle A rocket propelling payload from Earth into orbit.

Sounding Rocket A suborbital rocket for upper atmosphere experiments.

Spaceplane A vehicle operating in both atmosphere and space.

Hypersonic Vehicle A craft capable of sustained flight above Mach 5.

Flight Mechanics The study of forces and motion of aircraft in flight.

Sideslip Angle The angle between the aircraft axis and relative wind.

Flight Path The trajectory of the aircraft's center of gravity.

Glide Ratio Horizontal distance per altitude lost, equal to L/D.

Ceiling The maximum altitude an aircraft can reach.

Stall Recovery Techniques to regain control from a stalled condition.

Spin Recovery Techniques to recover from an autorotative spin.

Downwash Air deflected downward by a wing in producing lift.

Flow Separation Boundary layer detachment causing lift loss.

Fuselage The central body of an aircraft.

Wing Spar The principal spanwise structural member of a wing.

Rib A chordwise airfoil-shaped structural member of a wing.

Stringer A thin longitudinal structural reinforcing element.

Bulkhead A transverse partition carrying concentrated loads.

Longeron A major longitudinal fuselage member.

Empennage The tail assembly providing stability and control.

Vertical Stabilizer The fixed vertical tail surface for directional stability.

Horizontal Stabilizer The fixed horizontal tail surface for longitudinal stability.

Fin Another term for the vertical stabilizer.

Skin The outer covering transferring aerodynamic loads.

Honeycomb Panel A lightweight composite panel with a honeycomb core.

Monocoque A structure where the outer skin carries main stresses.

Semi-Monocoque A stressed skin structure with stringer and bulkhead reinforcement.

Aluminum Alloy Lightweight alloy extensively used in aircraft.

Titanium Alloy High-strength corrosion-resistant metal for high-temperature parts.

Carbon Fiber A strong, stiff, lightweight composite material.

Fiberglass Glass fibers embedded in polymer matrix.

Kevlar A strong heat-resistant synthetic fiber.

Composite Material A material combining two or more constituent materials.

Ceramic Matrix Composite A composite with ceramic matrix for high temperatures.

Superalloy An alloy with high strength and creep resistance at high temperatures.

Shape Memory Alloy An alloy returning to shape when heated after deformation.

Ablative Material A material eroding to carry away heat.

Hydraulic System Pressurized fluid system for actuating flight controls.

Pneumatic System Compressed air system for aircraft subsystems.

Electrical System The system providing electrical power to the aircraft.

Fuel System The system storing and delivering fuel to engines.

Environmental Control System The system managing cabin pressure, temperature, and ventilation.

Cabin Pressurization Maintaining safe cabin altitude at high flight levels.

Landing Gear The undercarriage supporting the aircraft on the ground.

Braking System Components decelerating the aircraft on the runway.

Anti-Skid System A safety system preventing wheel lock during braking.

Ice Protection System A system preventing ice on wings and tail.

Flight Computer The central computing unit for flight data.

Flight Management System An integrated computer for navigation and performance.

Autopilot A system automatically controlling the flight path.

Fly-by-Wire Electrical transmission of pilot commands.

Glass Cockpit A cockpit with electronic flight instrument displays.

Head-Up Display A transparent display showing flight data in the pilot's line of sight.

Multi-Function Display A configurable screen for navigation and systems data.

Air Data Computer A computer computing altitude, speed, and Mach number.

GPS Satellite-based global positioning system.

GNSS Global Navigation Satellite Systems: GPS, GLONASS, Galileo, BeiDou.

INS A self-contained navigation system using accelerometers and gyroscopes.

Dead Reckoning Position estimation from speed, time, and direction.

Waypoint A predetermined set of coordinates on a flight route.

Great Circle Route The shortest path between two points on a sphere.

Magnetic Heading Aircraft direction relative to magnetic north.

True Heading Aircraft direction relative to true north.

Indicated Airspeed Speed read directly from the airspeed indicator.

Calibrated Airspeed IAS corrected for instrument and position errors.

True Airspeed Airspeed relative to undisturbed air.

Ground Speed Speed relative to the ground.

Turbojet A jet engine passing all air through the core.

Turbofan A jet engine diverting some air around the core.

Turboprop A jet engine driving a propeller.

Turboshaft A jet engine optimized for shaft power.

Pulsejet A jet engine operating by intermittent combustion.

Combustion Chamber The engine section where fuel and air burn.

Afterburner A secondary combustion chamber for thrust augmentation.

Specific Impulse A measure of propellant efficiency in seconds.

Thrust Vector Control Redirecting engine thrust for vehicle control.

Payload The cargo delivered by a launch vehicle.

Fairing The nose cone protecting the payload during ascent.

Booster The first stage providing initial thrust.

Upper Stage The final stage completing orbit insertion.

Delta-v The velocity change capacity for orbital maneuvers.

Escape Velocity Speed needed to break free from gravity.

Orbital Velocity Speed required to stay in orbit.

Staging Discarding empty stages to reduce mass.

Gravity Turn Using gravity to pitch the vehicle during ascent.

Hot Staging Upper stage ignition before lower stage separation.

Spacecraft Bus The platform carrying the payload and supporting subsystems.

Solar Panel A device converting sunlight into electricity.

Reaction Wheel A flywheel for spacecraft attitude control.

Star Tracker An optical device determining attitude from star positions.

Sun Sensor A sensor detecting the Sun direction.

Magnetorquer A device using magnetic fields for torque.

Docking Port An interface for connecting two spacecraft.

Airlock A chamber maintaining pressure integrity during passage.

Heat Shield A layer dissipating re-entry thermal energy.

Orbit The curved path of an object around a celestial body.

Elliptical Orbit An orbit with eccentricity between 0 and 1.

Circular Orbit An orbit with eccentricity of 0.

Geostationary Orbit A circular orbit at 35,786 km synchronized to Earth's rotation.

Low Earth Orbit An orbit below 2,000 km.

Medium Earth Orbit An orbit between 2,000 and 35,786 km.

Polar Orbit An orbit passing over the Earth's poles.

Sun-Synchronous Orbit A polar orbit with fixed orientation relative to the Sun.

Transfer Orbit An orbit for moving between two orbital paths.

Hohmann Transfer An efficient elliptical transfer between circular orbits.

Orbital Inclination The angle between the orbital plane and the reference plane.

Eccentricity A measure of orbital deviation from circular.

Orbital Period Time for one complete revolution.

Guidance Determining the desired path for a vehicle.

Navigation Planning and controlling the vehicle course.

Attitude Control Controlling the orientation of a vehicle.

Attitude Determination Calculating vehicle orientation from sensor data.

PID Controller A feedback controller using proportional, integral, derivative terms.

Kalman Filter An algorithm estimating states from noisy measurements.

State Estimation Inferring system state from sensor data.

Sensor Fusion Combining multiple data sources for better information.

Control Law A mathematical equation determining control outputs.

Microgravity Near-weightlessness in free fall.

Space Vacuum Extremely low ambient pressure in space.

Space Weather Environmental conditions influenced by solar activity.

Solar Wind Charged particles streaming from the Sun.

Van Allen Belts Zones of energetic particles trapped by Earth's magnetic field.

Cosmic Ray High-energy particles from outer space.

Space Plasma Hot ionized gas in the magnetosphere and solar wind.

Space Debris Non-functional artificial objects in orbit.

Meteoroid A small rocky or metallic body in space.

Micrometeoroid A tiny meteoroid less than a millimeter.

Additive Manufacturing Building objects layer by layer.

CNC Machining Computer numerical controlled automated machining.

Friction Stir Welding A solid-state joining process using frictional heat.

Electron Beam Welding A welding process using high-velocity electrons.

Autoclave Curing Composite curing with heat and pressure in an autoclave.

Resin Transfer Molding A closed-mold composite manufacturing process.

Filament Winding Winding resin-impregnated fibers around a mandrel.

Fatigue Test Testing under repeated cyclic loads.

Flutter Test Testing for aeroelastic oscillation speeds.

Vibration Test Testing under controlled vibration conditions.

Thermal Vacuum Test Space hardware testing in simulated vacuum and thermal conditions.

Structural Test Physical testing of load-bearing components.

Flight Test Testing in actual flight conditions.

Qualification Test Tests demonstrating design requirements are met.

Acceptance Test Production unit tests verifying specifications.

Redundancy Backup components to increase reliability.

Fail-Safe Design Design preventing or mitigating failure consequences.

Fault Tolerance Continued operation after component failures.

MTBF Mean time between failures.

Reliability The ability to function under stated conditions.

Risk Assessment Systematic evaluation of possible risks.

Hazard Analysis Identification of potential hazards.

Flight Safety Ensuring safe aircraft or spacecraft operation.

Range Safety Protecting the public from launch vehicle hazards.

Airworthiness Suitability for safe flight.

Type Certificate Official approval of an aircraft design.

Supplemental Type Certificate Approval of major modifications to a certified design.

FAA Federal Aviation Administration, US aviation authority.

EASA European Union Aviation Safety Agency.

ICAO International Civil Aviation Organization.

DO-178C Guidelines for aeronautical software development.

DO-254 Guidelines for complex aeronautical hardware development.

Urban Air Mobility Air transportation within cities.

Electric Aircraft Aircraft powered by electric motors and batteries.

Hybrid-Electric Propulsion Combining conventional and electric propulsion.

Hydrogen Aircraft Aircraft using hydrogen as fuel.

Sustainable Aviation Fuel Fuel from non-fossil sources.

Autonomous Flight Flight without human intervention.

Digital Twin A virtual model of a physical system.

Morphing Wing A wing changing shape in flight.

Reusable Launch Vehicle A rocket with recoverable stages.

Cockpit The compartment from which the pilot controls the aircraft.

Canopy The transparent enclosure over the cockpit.

Nose Gear The forward landing gear.

Main Gear The primary landing gear supporting most weight.

Tailwheel A rear wheel on conventional landing gear.

Tricycle Gear Landing gear with nose wheel and two main wheels.

Brake A device for slowing or stopping the aircraft.

Parking Brake A brake holding the aircraft stationary.

Control Column The primary flight control input device.

Yoke The control wheel for pitch and roll.

Cyclic The control stick for helicopter rotor tilt.

Collective The control for helicopter rotor pitch.

Throttle The power control for the engine.

Mixture Control Fuel-air mixture adjustment for piston engines.

Propeller Control RPM and pitch control for constant-speed propellers.

Trim Tab A small adjustable surface for reducing control forces.

Balance Tab A tab that moves opposite to the control surface.

Anti-Servo Tab A tab moving with the control surface for feel.

Mass Balance Weights counterbalancing control surfaces.

Aerodynamic Balance A surface shape reducing control forces.

Horn Balance An extension forward of the hinge line.

Internal Balance A sealed balance chamber within the control surface.

Boosted Controls Power-assisted flight controls.

Power Controls Fully powered flight controls.

Control Horn An arm on a control surface for linkage attachment.

Pushrod A rod transmitting control input mechanically.

Cable A wire cable for control transmission.

Bellcrank A lever changing the direction of control movement.

Quadrant A sector-shaped lever for control input.

Linkage The mechanical connection system for controls.

Control Surface A movable surface for aerodynamic control.

Primary Control Main control surfaces for pitch, roll, and yaw.

Secondary Control Auxiliary surfaces like flaps and trim tabs.

High-Lift Device A device increasing lift at low speeds.

Leading Edge Device A device on the leading edge for high lift.

Trailing Edge Device A device on the trailing edge for high lift.

Fowler Flap A flap extending aft and down for increased area and camber.

Split Flap A flap deflecting from the lower surface only.

Slotted Flap A flap with a gap for high-energy air to the upper surface.

Krueger Flap A leading edge flap hinged at the front.

Handley Page Slat A leading edge slat that extends forward.

Blown Flap A flap with engine bleed air for boundary layer control.

Jet Flap A flap using engine exhaust for lift augmentation.

Circulation Control Wing A wing using blowing for lift control.

Active Flow Control Active methods for manipulating flow.

Load Alleviation Reducing loads through active control.

Gust Alleviation Active control reducing gust loads.

Mode Control Panel The panel for autopilot mode selection.

Navigation Display A display showing navigation information.

Primary Flight Display The main display showing attitude and airspeed.

Engine Display A display showing engine parameters.

System Display A display showing aircraft systems status.

Caution Warning System A system alerting the crew to abnormal conditions.

Master Warning A primary alert indicator.

EICAS Engine Indicating and Crew Alerting System.

ECAM Electronic Centralized Aircraft Monitor.

Data Bus A communication network for aircraft systems.

ARINC 429 A standard for avionics data communication.

MIL-STD-1553 A military standard for data bus communication.

Aircraft Wiring Electrical wiring in aircraft systems.

Circuit Breaker A resettable overcurrent protection device.

Bus Bar A power distribution conductor.

Generator A device converting mechanical to electrical power.

Alternator An AC generator in aircraft electrical systems.

Battery An electrochemical power storage device.

APU Auxiliary Power Unit, a small engine for ground power.

Inverter A device converting DC to AC power.

Transformer A device changing AC voltage levels.

Rectifier A device converting AC to DC.

Voltage Regulator A device maintaining constant voltage.

Oxygen System A system providing supplemental oxygen.

Pressurization System A system maintaining cabin pressure.

Air Conditioning A system controlling cabin temperature.

Bleed Air Compressed air from the engine for pneumatic systems.

ECS Pack An environmental control system unit.

Ventilation Cabin air circulation system.

Rotor Brake A brake stopping the rotor after shutdown.

Rotor Head The assembly connecting rotor blades to the mast.

Swashplate A mechanism translating controls to blade pitch.

Collective Pitch Simultaneous pitch change of all rotor blades.

Cyclic Pitch Cyclic variation of blade pitch for rotor tilt.

Lead-Lag Hinge A hinge allowing blade fore-aft movement.

Flapping Hinge A hinge allowing blade vertical movement.

Feathering Hinge A hinge allowing blade pitch change.

Fully Articulated Rotor A rotor with all three hinges.

Semi-Rigid Rotor A rotor with a teetering hinge.

Rigid Rotor A rotor without hinges, relying on blade flexibility.

Tail Rotor A rotor counteracting main rotor torque.

Fenestron A ducted tail rotor.

NOTAR No tail rotor, using circulation control.

Main Rotor The primary lifting rotor of a helicopter.

Tip Speed The speed of the rotor blade tip.

Hover Flight at zero ground speed.

Autorotation Landing A landing without engine power.

Ground Resonance A destructive oscillation of rotor and airframe.

Retreating Blade Stall Stall of the retreating rotor blade at high speed.

Dissymmetry of Lift Unequal lift between advancing and retreating blades.

Translational Lift Lift increase with forward speed.

Effective Translational Lift Lift increase as the rotor leaves its downwash.

Ground Effect Helicopter Increased lift near the ground.

Vortex Ring State A dangerous descent condition for helicopters.

Settling With Power Another term for vortex ring state.

Power Required Curve A curve showing power needed for flight.

Power Available Curve A curve showing power available from the engine.

Height-Velocity Diagram A diagram showing safe hover and landing regions.

H-V Curve The height-velocity diagram for helicopters.

Cat A Takeoff A takeoff with one engine failure capability.

Cat B Takeoff A takeoff assuming no engine failure.

Stall Warning System A system alerting the pilot of an impending stall condition.

Stick Shaker A device vibrating the control column to indicate an approaching stall.

Yaw Damper An automatic flight control system damping Dutch roll oscillations.

Autothrottle An automatic system controlling engine thrust to maintain speed.

Flight Director A flight instrument displaying command guidance cues to the pilot.

TCAS Traffic Alert and Collision Avoidance System for preventing mid-air collisions.

GPWS Ground Proximity Warning System alerting the crew of terrain hazards.

Radar Altimeter An instrument measuring altitude using radio wave reflection.

Transponder An aircraft transceiver for ATC identification and altitude reporting.

Rockets and Missiles

Rocket propulsion, launch vehicle design, and guided missile systems. From the Tsiolkovsky equation to terminal guidance and hypersonic glide vehicles.

Rocket Engine A thrust chamber where propellant is burned or heated and expelled.

Solid Rocket Motor A rocket using solid chemical propellant cast in a case.

Liquid Rocket Engine An engine using liquid fuel and oxidizer pumped to the combustion chamber.

Hybrid Rocket A rocket combining solid fuel with liquid or gaseous oxidizer.

Specific Impulse A measure of rocket propellant efficiency.

Total Impulse The total thrust integrated over burn time.

Thrust The force propelling a rocket forward.

Thrust-to-Weight Ratio The ratio of engine thrust to vehicle weight.

Propellant The fuel and oxidizer mixture burned in a rocket engine.

Oxidizer The substance supplying oxygen for combustion in a rocket.

Bipropellant A rocket using separate fuel and oxidizer.

Monopropellant A single propellant decomposed by a catalyst.

Hypergolic A propellant combination that ignites on contact.

Cryogenic Propellant Propellant stored at extremely low temperatures.

RP-1 A refined kerosene used as rocket fuel.

Liquid Hydrogen A cryogenic fuel used in high-performance rockets.

Liquid Oxygen A cryogenic oxidizer used with hydrogen and kerosene.

Nitrogen Tetroxide A storable oxidizer for hypergolic propellants.

Hydrazine A storable monopropellant and fuel.

MMH Monomethylhydrazine, a hypergolic fuel.

UDMH Unsymmetrical dimethylhydrazine, a hypergolic fuel.

Propellant Tank A tank storing rocket propellant.

Tank Pressurization Pressurizing tanks to feed propellant to the engine.

Combustion Chamber The chamber where propellant burns in a rocket engine.

Injector A device mixing fuel and oxidizer in the combustion chamber.

Nozzle A duct accelerating exhaust gases to produce thrust.

De Laval Nozzle A convergent-divergent nozzle for supersonic exhaust.

Throat The narrowest section of a rocket nozzle.

Expansion Ratio The ratio of nozzle exit area to throat area.

Nozzle Contour The shape of a rocket nozzle wall.

Rao Nozzle A nozzle contoured for maximum thrust.

Bell Nozzle A curved contour nozzle for high altitude.

Plug Nozzle An altitude-compensating nozzle design.

Aerospike Nozzle A nozzle using external flow for altitude compensation.

Regenerative Cooling Cooling a nozzle by circulating propellant through it.

Film Cooling Cooling by injecting a coolant film on the chamber wall.

Ablative Cooling Cooling by ablating the inner surface material.

Radiative Cooling Cooling by thermal radiation from the nozzle.

Igniter A device initiating combustion in the chamber.

Pyrotechnic Igniter An igniter using pyrotechnic compounds.

Spark Igniter An igniter using an electric spark.

Catalytic Igniter An igniter using a catalyst for hypergolic or monopropellant ignition.

Thrust Chamber The assembly including combustion chamber and nozzle.

Engine Cycle The propellant feed cycle of a liquid rocket engine.

Pressure Fed Cycle An engine cycle using tank pressure for propellant feed.

Gas Generator Cycle A cycle burning a small portion of propellant to drive a turbine.

Staged Combustion Cycle A cycle burning preburner exhaust in the main chamber.

Full Flow Staged Combustion A staged combustion cycle with two preburners.

Expander Cycle A cycle using heat absorption to drive the turbine.

Tap-Off Cycle A cycle tapping combustion gas to drive the turbine.

Electric Pump Cycle A cycle using electric motors to drive pumps.

Turbopump A turbine-driven pump feeding propellant to the engine.

Preburner A small chamber generating gas to drive the turbine.

Main Combustion Chamber The primary chamber where full combustion occurs.

Thrust Vector Control Steering a rocket by redirecting thrust.

Gimbal A swivel mount for engine thrust vectoring.

Vernier Engine A small engine for fine trajectory adjustments.

Reaction Control System Small thrusters for attitude control in space.

Cold Gas Thruster A thruster using compressed gas for attitude control.

Launch Vehicle A rocket used to launch payloads into space.

Expendable Launch Vehicle A single-use launch vehicle.

Reusable Launch Vehicle A launch vehicle with recoverable stages.

Multi-Stage Rocket A rocket with multiple stages discarded during ascent.

First Stage The initial stage providing lift-off thrust.

Second Stage The stage igniting after first stage separation.

Upper Stage The final stage for orbit insertion.

Strap-On Booster An auxiliary booster attached to the first stage.

Payload Fairing The nose cone protecting the payload.

Interstage A structure connecting rocket stages.

Stage Separation The process of discarding an expended stage.

Fairing Separation Jettisoning the payload fairing after atmospheric exit.

Launch Pad The facility from which a rocket launches.

Launch Tower A structure supporting the rocket before launch.

Launch Rail A rail guiding the rocket during initial ascent.

Launch Tube A tube for launching guided missiles.

Mobile Launcher A transportable launch platform.

Silo An underground launch facility for missiles.

Countdown The timeline leading to launch.

T-0 The moment of engine ignition or lift-off.

Lift-Off The moment the rocket leaves the ground.

Max Q The point of maximum dynamic pressure during ascent.

MECO Main engine cutoff.

SECO Second engine cutoff.

Insertion The process of entering the intended orbit.

Injection Burn A burn to inject into a transfer trajectory.

Circularization Burn A burn to circularize an orbit.

Deorbit Burn A burn to initiate return to Earth.

Retro Rocket A rocket firing opposite to the direction of travel.

Ullage Motor A small motor settling propellant before main engine ignition.

Spin Motor A motor spinning a stage for stability.

Separation Motor A small motor separating stages.

IRFNA Inhibited red fuming nitric acid, a storable oxidizer.

LOX Liquid oxygen.

LH2 Liquid hydrogen.

LNG Liquefied natural gas as rocket fuel.

Propellant Mass Fraction The ratio of propellant mass to total mass.

Dry Mass The mass of the rocket without propellant.

Wet Mass The mass of the rocket with propellant.

Payload Mass Fraction The ratio of payload mass to total mass.

Tsiolkovsky Equation The rocket equation: delta-v = ve * ln(m0/mf).

Characteristic Velocity A measure of combustion performance (c*).

Exhaust Velocity The speed of exhaust gases leaving the nozzle.

Effective Exhaust Velocity The effective speed of exhaust including pressure effects.

Mass Flow Rate The mass of propellant flowing per unit time.

Chamber Pressure The pressure inside the combustion chamber.

Expansion Ratio Effect The effect of nozzle expansion on performance.

Optimal Expansion Nozzle expansion matching ambient pressure.

Overexpansion Nozzle exit pressure below ambient.

Underexpansion Nozzle exit pressure above ambient.

Flow Separation in Nozzle Detachment of flow from the nozzle wall.

Side Load Lateral loads from flow separation in a nozzle.

Nozzle Extension An extendable nozzle section for altitude adaptation.

Dual Bell Nozzle A nozzle with two bell contours for altitude compensation.

Thrust Chamber Liner A liner protecting the thrust chamber wall.

Combustion Stability Stable combustion without pressure oscillations.

Combustion Instability Pressure oscillations in the combustion chamber.

Chugging Low-frequency combustion instability.

Buzzing Intermediate frequency instability.

Screaming High-frequency combustion instability.

Pogo Oscillation Longitudinal oscillation from engine-vehicle interaction.

Acoustic Cavity A cavity damping acoustic modes in the chamber.

Baffle A device in the injector to prevent instability.

Resonator A device damping specific frequencies.

Injector Pattern The arrangement of injector elements.

Showerhead Injector A simple injector with many small holes.

Impinging Injector An injector where jets impinge for mixing.

Coaxial Injector An injector with concentric fuel and oxidizer streams.

Pintle Injector An injector with a central pintle for variable flow.

Shear Injector An injector using shear for atomization.

Atomization The breakup of propellant into fine droplets.

Droplet Combustion Burning of individual propellant droplets.

Combustion Efficiency How completely propellant is burned.

C Star Efficiency Combustion efficiency measured by characteristic velocity.

Thrust Coefficient A dimensionless measure of nozzle performance.

Specific Impulse Efficiency The achieved Isp as a fraction of theoretical.

Grain A shaped solid propellant charge.

Grain Configuration The geometric shape of the solid grain.

Burning Surface The surface area of the grain that is burning.

Neutral Burning Burning with constant surface area.

Progressive Burning Burning with increasing surface area.

Regressive Burning Burning with decreasing surface area.

End Burning Burning from one end of the grain.

Internal Burning Burning from internal passages.

Cigarette Burning End burning of a solid grain.

Star Grain A grain with a star-shaped internal bore.

Wagon Wheel Grain A grain with a wagon wheel cross-section.

Slotted Tube Grain A tubular grain with slots for burning surface control.

Inhibitor A coating preventing burning on certain grain surfaces.

Propellant Liner A liner bonding the grain to the case.

Case The pressure vessel containing the solid propellant.

Motor Case The structural casing of a solid rocket motor.

Insulation Thermal protection inside the motor case.

Nozzle Throat Insert A high-temperature material at the nozzle throat.

Erosion Material loss from the nozzle throat during firing.

Graphite Nozzle A nozzle using graphite for high temperature resistance.

Carbon-Carbon Nozzle A nozzle using carbon-carbon composite.

Metal Nozzle A nozzle made of high-temperature metal.

Nozzle Closure A seal protecting the nozzle before ignition.

Igniter Grain A small grain for motor ignition.

Pyrogen Igniter A pyrotechnic igniter generating hot gas.

Safe and Arm Device A safety device preventing accidental ignition.

Thrust Termination A system to shut down thrust.

Destruct System A system destroying the rocket in flight.

Flight Termination System A system for range safety destruct.

Range Safety Protecting the public from launch hazards.

Missile A guided weapon system.

Ballistic Missile A missile following a ballistic trajectory.

Cruise Missile A self-navigating missile with continuous thrust.

ICBM An intercontinental ballistic missile.

IRBM An intermediate-range ballistic missile.

SRBM A short-range ballistic missile.

SLBM A submarine-launched ballistic missile.

SAM A surface-to-air missile.

AAM An air-to-air missile.

ASM An air-to-surface missile.

ATGM An anti-tank guided missile.

Anti-Ship Missile A missile targeting ships.

Hypersonic Missile A missile capable of speeds above Mach 5.

Hypersonic Glide Vehicle A maneuverable hypersonic boost-glide weapon.

MIRV Multiple independently targetable reentry vehicles.

Reentry Vehicle The part of a missile carrying the warhead through reentry.

Warhead The explosive or payload of a missile.

Conventional Warhead A non-nuclear explosive warhead.

Nuclear Warhead A nuclear explosive warhead.

Guidance System The system directing a missile to its target.

Inertial Guidance Guidance using accelerometers and gyroscopes.

GPS Guidance Guidance using satellite positioning.

Terrain Contour Matching Guidance matching terrain data.

Scene Matching Guidance matching imagery data.

Active Radar Homing Homing using onboard radar transmitter and receiver.

Semi-Active Radar Homing Homing using reflected radar from an external illuminator.

Passive Radar Homing Homing using emissions from the target.

Infrared Homing Homing using heat emissions from the target.

Laser Homing Homing using reflected laser energy.

Command Guidance Guidance from an external command link.

Beam Riding Guidance following a directed beam.

Wire Guidance Guidance via wires spooled from the missile.

Fiber Optic Guidance Guidance via fiber optic link.

Terminal Guidance The final phase of missile guidance.

Midcourse Guidance The intermediate phase of missile guidance.

Boost Phase The initial phase of missile flight under power.

Coast Phase The unpowered phase of missile flight.

Reentry Phase The final phase returning through the atmosphere.

Trajectory The path followed by a missile or rocket.

Ballistic Trajectory A trajectory governed by gravity and inertia.

Lifting Trajectory A trajectory using aerodynamic lift.

Depressed Trajectory A flatter than normal ballistic trajectory.

Lofted Trajectory A higher than normal ballistic trajectory.

Minimum Energy Trajectory The most efficient trajectory for a given range.

Maximum Range Trajectory The trajectory achieving the greatest range.

Launch Azimuth The direction of launch measured from north.

Impact Point The point where the missile strikes.

CEP Circular error probable, a measure of accuracy.

Kill Probability The probability of destroying the target.

Lethality The damaging effect of a warhead.

Fragmentation Warhead A warhead releasing fragments.

Shaped Charge A shaped explosive for armor penetration.

Kinetic Kill Vehicle A hit-to-kill interceptor.

Decoy A device misleading enemy defenses.

Countermeasure A system defeating enemy targeting.

Chaff Radar-reflective material for decoying.

Flare A heat source decoying infrared seekers.

Electronic Countermeasure Electronic jamming or deception.

Penetration Aid A device helping a missile penetrate defenses.

Anti-Ballistic Missile A missile intercepting ballistic missiles.

Interceptor A missile designed to destroy incoming threats.

Hit-to-Kill Direct impact interception.

Proximity Fuse A fuse detonating near the target.

Contact Fuse A fuse detonating on impact.

Time Fuse A fuse detonating after a set time.

Airburst An explosion in the air above the target.

Terminal Phase The final engagement phase of an interceptor.

Launch Detection Detecting a missile launch by satellites or radar.

Tracking Radar Radar following a target.

Fire Control Radar Radar directing weapons.

Phased Array Radar A radar with electronic beam steering.

Early Warning Radar Radar providing early threat detection.

Over-the-Horizon Radar Radar seeing beyond the horizon.

Satellite Warning System A satellite system detecting missile launches.

Ballistic Missile Defense A system defending against ballistic missiles.

THAAD Terminal High Altitude Area Defense.

Patriot A US surface-to-air missile system.

Aegis A US naval combat system for missile defense.

Ground-Based Interceptor A US ground-based missile interceptor.

Iron Dome An Israeli short-range air defense system.

David Sling An Israeli medium-range air defense system.

Arrow An Israeli anti-ballistic missile system.

S-400 A Russian long-range air defense system.

S-500 A Russian next-generation air defense system.

Ramjet Missile A missile using a ramjet engine.

Scramjet Missile A missile using a scramjet engine.

Solid Rocket Booster A solid rocket used as a strap-on booster.

Dual Thrust Motor A motor with two thrust levels.

Boost-Sustain Motor A motor providing boost then sustained thrust.

Thrust Profile The variation of thrust over burn time.

Flame Deflector A structure deflecting exhaust at launch.

Exhaust Plume The visible exhaust of a rocket engine.

Mach Diamond Diamond-shaped shock patterns in exhaust.

Sounding Rocket A suborbital rocket for research.

Meteorological Rocket A rocket for atmospheric measurements.

Target Drone An unmanned target used for missile testing.

Burn Rate The rate at which solid propellant burns.

Strand Burn Rate Burn rate measured on a small sample.

Exhaust Trail The visible trail from a missile exhaust.

Liquid Injection Thrust Vectoring Injection of liquid into the nozzle for thrust vectoring.

Jet Vanes Vanes in the exhaust for steering.

Jet Tab A tab in the exhaust for steering.

Flexible Bearing A flexible joint for nozzle gimballing.

Hot Gas Valve A valve directing hot gas for thrust vectoring.

Secondary Injection Injection of secondary fluid for thrust vectoring.

Main Propellant Valve The valve controlling propellant flow to the engine.

Fill Valve A valve for filling propellant tanks.

Drain Valve A valve for draining propellant.

Vent Valve A valve for tank venting.

Relief Valve A valve preventing overpressure.

Check Valve A valve allowing flow in one direction only.

Solenoid Valve An electrically actuated valve.

Poppet Valve A valve with a disc and seat.

Ball Valve A valve with a rotating ball.

Butterfly Valve A valve with a rotating disc.

Pneumatic Valve A valve actuated by compressed gas.

Pyro Valve A valve actuated by pyrotechnic charge.

Propellant Line A line carrying propellant.

Suction Line A line on the inlet side of the pump.

Discharge Line A line on the outlet side of the pump.

Helium Pressurization Pressurization using helium.

Nitrogen Pressurization Pressurization using nitrogen.

Autogenous Pressurization Pressurization using vaporized propellant.

Accumulator A pressure storage reservoir.

Pressure Regulator A device maintaining constant pressure.

Gas Generator A device generating gas to drive a turbine.

Heat Exchanger A device transferring heat between fluids.

Gas Cooler A device cooling gas before use.

Filter A device removing particles from propellant.

Propellant Management The system managing propellant in tanks.

Propellant Settling Using small thrusters to settle propellant.

Fin-Stabilized Rocket A rocket stabilized by aerodynamic fins mounted on the rear.

Spin-Stabilized Rocket A rocket stabilized by rotating around its longitudinal axis.

Tail Fins Aerodynamic surfaces on the rear of a rocket for stability.

Canard Control Forward-mounted control surfaces for missile maneuvering.

Drones

Unmanned aerial vehicles: multirotor and fixed-wing drones, flight controllers, autonomy, regulations, and the technology powering the UAV revolution.

Drone An unmanned aerial vehicle operated remotely or autonomously.

UAV Unmanned Aerial Vehicle, the aircraft component of a UAS.

UAS Unmanned Aircraft System including vehicle, ground control, and data link.

RPAS Remotely Piloted Aircraft System.

Multirotor A drone with multiple rotors, typically 4, 6, or 8.

Quadcopter A drone with four rotors arranged in a cross configuration.

Hexacopter A drone with six rotors for redundancy and lift capacity.

Octocopter A drone with eight rotors for maximum redundancy.

Fixed-Wing Drone A drone with a rigid wing for efficient forward flight.

VTOL Drone A drone capable of vertical takeoff and landing with fixed-wing efficiency.

Hybrid Drone A drone combining multirotor and fixed-wing configurations.

Nano Drone A very small drone, typically under 250 grams.

Micro Drone A small drone, usually under 2 kg.

Tactical Drone A mid-range drone for battlefield or operational use.

MALE Drone Medium Altitude Long Endurance drone between 10,000 and 30,000 feet.

HALE Drone High Altitude Long Endurance drone above 30,000 feet.

Combat Drone Armed drone designed for military strike missions.

Reconnaissance Drone A drone designed for intelligence gathering.

Swarm Drone A drone designed to operate as part of a coordinated group.

Delivery Drone A drone designed for parcel and cargo delivery.

Agriculture Drone A drone optimized for crop monitoring and spraying.

Surveying Drone A drone equipped for mapping and land surveying.

Inspection Drone A drone used for infrastructure inspection.

Search and Rescue Drone A drone with thermal cameras and sensors for SAR operations.

Racing Drone A high-speed drone designed for competitive FPV racing.

FPV Drone First Person View drone piloted via onboard camera feed.

Cinematic Drone A drone with high-quality cameras for filmmaking.

Submersible Drone An underwater drone for marine inspection.

Autonomous Drone A drone capable of self-navigation without pilot input.

Tethered Drone A drone powered via a cable for unlimited endurance.

Solar Drone A drone powered by solar panels.

Flight Controller The onboard computer managing drone flight dynamics and stability.

IMU Inertial Measurement Unit combining accelerometers and gyroscopes.

Accelerometer A sensor measuring linear acceleration along three axes.

Gyroscope A sensor measuring angular velocity about three axes.

Magnetometer A sensor measuring magnetic field for heading reference.

Barometer A sensor measuring atmospheric pressure for altitude hold.

GPS Module A receiver providing global position data for navigation.

GLONASS Russian satellite navigation system used alongside GPS.

Compass A magnetic heading sensor for drone orientation.

ESC Electronic Speed Controller driving the drone's motors.

Motor The electric motor powering a drone rotor.

Brushless Motor An efficient, durable motor commonly used in drones.

Propeller The rotating blade producing thrust on a drone.

Pitch Propeller A propeller designed for clockwise rotation.

Yaw Propeller Counter-rotating propellers cancel torque for stable flight.

LiPo Battery Lithium Polymer battery used for drone power.

Battery Capacity Battery energy measured in milliampere-hours (mAh).

S Rating Number of cells in series in a LiPo battery.

C Rating Discharge rate rating of a LiPo battery.

Power Distribution Board A board distributing battery power to the ESCs.

PDB Power Distribution Board.

BEC Battery Eliminator Circuit providing regulated voltage.

Telemetry Real-time data transmission from drone to ground station.

Receiver The radio receiver on the drone for pilot commands.

Transmitter The radio transmitter used by the pilot to control the drone.

Radio Link The communication channel between transmitter and drone.

Control Link The uplink for sending commands to the drone.

Video Link The downlink transmitting camera video to the ground.

Data Link A bi-directional link for telemetry and commands.

Frequency Band The radio frequency used for drone communication.

2.4 GHz Common control frequency band for RC systems.

5.8 GHz Common frequency band for FPV video transmission.

900 MHz A longer-range frequency band for telemetry.

433 MHz An ultra-long-range frequency band.

FHSS Frequency Hopping Spread Spectrum, a robust modulation method.

DSSS Direct Sequence Spread Spectrum modulation.

LBT Listen Before Talk, a spectrum access protocol.

RSSI Received Signal Strength Indicator for link quality.

Antenna A device transmitting or receiving radio signals.

Omnidirectional Antenna An antenna radiating equally in all directions.

Directional Antenna An antenna focusing signal in one direction.

Patch Antenna A flat directional antenna.

Yagi Antenna A high-gain directional antenna.

Gimbal A stabilizing mount for cameras and sensors.

3-Axis Gimbal A gimbal stabilizing pitch, roll, and yaw.

2-Axis Gimbal A gimbal stabilizing pitch and roll.

PID Tuning Adjusting PID controller gains for optimal flight performance.

Autotune Automatic PID tuning by the flight controller.

Rate Mode A flight mode controlling angular rates directly.

Angle Mode A flight mode limiting tilt angle for stable flight.

Horizon Mode A flight mode with both rate and angle characteristics.

Altitude Hold A mode maintaining a constant barometric altitude.

Position Hold A mode maintaining GPS position and altitude.

Return to Home A mode automatically flying the drone back to its launch point.

RTH Return to Home.

Loiter A mode holding position and altitude.

Waypoint Navigation Autonomous flight through programmed GPS waypoints.

Follow Me A mode following a GPS-tracked person or device.

Orbit Mode A mode circling around a point of interest.

Mission Planning Creating an autonomous flight plan with waypoints and actions.

Geofence A virtual boundary restricting the drone's flight area.

Low Battery Warning An alert when battery voltage drops below a threshold.

Collision Avoidance A system detecting and avoiding obstacles.

Obstacle Detection Sensing obstacles using cameras, LiDAR, or ultrasonic sensors.

Emergency Stop An immediate motor kill command.

Lost Link Loss of communication between the transmitter and drone.

Failsafe An automatic action when a critical condition occurs.

Flight Termination System A system ending the flight, typically by cutting power.

Parachute System A recovery parachute for emergency descent.

VLOS Operating a drone within the pilot's unaided visual range.

BVLOS Operating a drone beyond the pilot's visual range.

EVLOS Extended Visual Line of Sight using observers.

Remote ID A system broadcasting drone identity and location.

No-Fly Zone A restricted area where drone flight is prohibited.

Airspace The controlled or uncontrolled airspace in which drones operate.

UTM Unmanned Aircraft System Traffic Management.

NOTAM Notice to Air Missions, an airspace advisory.

Flight Authorization Permission required to fly in controlled airspace.

LAANC Low Altitude Authorization and Notification Capability.

Part 107 FAA regulations for small UAS commercial operations.

Drone Registration Registering a drone with the aviation authority.

Operator License A certification required to operate a drone commercially.

Remote Pilot Certificate A certificate for commercial drone pilots.

Aerial Photography Capturing images or video from a camera mounted on a drone.

Precision Agriculture Using drones for crop monitoring and management.

Crop Monitoring Aerial surveillance of crop health and growth.

Mapping Creating orthophoto maps from drone imagery.

Surveying Measuring land features using drone photogrammetry.

Photogrammetry Creating 3D models from overlapping drone photographs.

Infrastructure Inspection Inspecting bridges, power lines, and structures with drones.

Search and Rescue Using drones to locate missing persons.

Disaster Response Using drones for damage assessment and relief coordination.

Wildlife Monitoring Tracking and counting wildlife populations with drones.

Parcel Delivery Delivering packages using autonomous drones.

Environmental Monitoring Tracking environmental parameters with drone sensors.

Mission Planner A ground station software for planning autonomous missions.

QGroundControl An open-source ground control station software.

PX4 An open-source flight control software for drones.

ArduPilot An open-source autopilot software suite.

MAVLink A lightweight communication protocol for drones.

MAVSDK A SDK for MAVLink-based drone control.

Gazebo A 3D robotics simulator used for drone simulation.

AirSim A drone and car simulator from Microsoft.

SITL Software-in-the-Loop simulation running the autopilot natively.

HITL Hardware-in-the-Loop simulation with actual flight hardware.

Digital Twin A virtual replica of a physical drone system.

ROS Robot Operating System used for drone software development.

DroneKit A Python library for drone control.

Camera Payload A camera mounted on a drone for imaging.

LiDAR Payload A laser scanning system for 3D mapping.

Sprayer A liquid spray system for agricultural drones.

Cargo Pod A container for carrying goods on a delivery drone.

Loudspeaker An audio system for public address from a drone.

Spotlight A high-intensity light mounted on a drone.

Winch A hoisting mechanism for cargo delivery.

Delivery Mechanism A system for releasing cargo from a drone.

Scientific Instrument A specialized sensor for scientific measurements.

Hyperspectral Sensor A sensor capturing many spectral bands.

Thermal Camera A camera sensing infrared heat signatures.

Multispectral Sensor A sensor capturing specific wavelength bands.

Flight Time The maximum duration a drone can stay airborne.

Hover Time The maximum time a drone can maintain a stationary hover.

Endurance The maximum time a drone can stay in the air.

Range The maximum distance a drone can fly from its controller.

Payload Capacity The maximum weight a drone can carry.

MTOM Maximum Takeoff Mass.

MTOW Maximum Takeoff Weight.

Service Ceiling The maximum altitude a drone can maintain.

Maximum Speed The highest speed the drone can achieve.

Cruise Speed The most efficient speed for drone flight.

Wind Resistance The maximum wind speed in which a drone can operate safely.

IP Rating Ingress Protection rating for environmental sealing.

Drone Swarm A group of drones operating as a coordinated entity.

Swarm Intelligence Collective behavior emerging from local interactions.

Formation Flight Drones flying in a structured formation.

Cooperative Navigation Multiple drones sharing sensor data for navigation.

Distributed Control Control decisions distributed among swarm members.

Mesh Network A decentralized communication network between drones.

Multi-Agent System A system of multiple autonomous agents working together.

Consensus Algorithm An algorithm for reaching agreement in a distributed system.

Swarm Coordination Managing the actions of multiple drones to achieve a goal.

Collision Avoidance Swarm Avoiding collisions between drones in a swarm.

FAA Federal Aviation Administration, US drone regulator.

EASA European Union Aviation Safety Agency.

ICAO International Civil Aviation Organization.

ASTM International An international standards organization for drones.

RTCA An organization developing aviation standards.

Dronecode Foundation An organization supporting open-source drone software.

PX4 Community The community developing the PX4 autopilot.

ArduPilot Project The community developing the ArduPilot software.

Flying Weight The total weight of the drone at takeoff.

Frame The structural chassis of a drone.

Arm A structural extension carrying a motor on a multirotor.

Landing Skid The landing support of a drone.

Motor Mount The bracket attaching the motor to the arm.

Propeller Guard A protective ring around the propeller.

Camera Mount The bracket attaching the camera to the drone.

Vibration Damping Isolating sensors from motor vibrations.

Anti-Vibration Mount A mount reducing vibration transmission.

Foam Damping Foam material used for vibration isolation.

GPS Mast A raised mount for the GPS antenna.

Compass Mount A mount isolating the compass from magnetic interference.

LED Indicator A light showing drone status.

Buzzer An audible indicator for locating the drone.

Current Sensor A sensor measuring battery current draw.

Voltage Sensor A sensor measuring battery voltage.

Power Module A module providing voltage and current sensing.

Pixhawk A popular open-source flight controller hardware.

Cube A modular flight controller system.

Kakute A flight controller series for racing drones.

F4 Processor An STM32F4-based flight controller processor.

F7 Processor An STM32F7-based flight controller processor.

H7 Processor An STM32H7-based flight controller processor.

Betaflight A popular firmware for racing and freestyle drones.

INAV An open-source firmware for navigation and autonomous flight.

Cleanflight An early open-source drone firmware.

Kiss A commercial flight controller firmware.

ButterFlight A fork of Betaflight for smooth flight.

EmuFlight A Betaflight fork for improved performance.

Raceflight A firmware for racing drones.

ExpressLRS An open-source RC link protocol.

Crossfire A long-range RC link from TBS.

Tracer A low latency RC link from TBS.

Ghost An ultra-low latency RC link.

DSMX Spektrum's DSM protocol variant.

SBUS A serial bus protocol for RC receivers.

CRSF Crossfire serial protocol.

PWM Pulse Width Modulation for servo/motor control.

DShot A digital ESC protocol.

Multishot An ESC protocol with higher resolution.

Oneshot An early high-frequency ESC protocol.

Bidirectional DShot DShot with telemetry feedback.

RPM Filtering Using motor RPM for dynamic notch filtering.

Dynamic Filter An adaptive filter removing noise frequencies.

Notch Filter A filter removing a specific frequency.

Low Pass Filter A filter passing low frequencies and attenuating high.

Gyro Sampling The rate at which the gyroscope is read.

PID Loop Frequency The frequency of the PID control loop.

Desync Loss of synchronization between ESC and motor.

Motor Timing The timing advance of motor commutation.

Demag Compensation Compensation for motor back EMF during commutation.

Throttle Scale Scaling the throttle output for linear response.

Throttle Limit A limit on maximum throttle output.

Motor Output The signal sent by the FC to the ESC.

Direction Lock Fixing motor rotation direction in software.

Resource Remapping Reassigning pin functions on the flight controller.

Serial Port A communication port for peripherals.

I2C An inter-integrated circuit communication bus.

SPI Serial Peripheral Interface bus.

UART Universal Asynchronous Receiver-Transmitter.

CAN Bus Controller Area Network bus.

SD Card Logging Logging flight data to an SD card.

Blackbox Flight data logging on a drone.

OSD On-Screen Display showing flight data in the FPV feed.

Betaflight OSD The OSD system in Betaflight firmware.

DJI OSD The OSD in DJI's FPV system.

FPV Camera A camera transmitting live video to the pilot's goggles.

VTX Video Transmitter for FPV.

Video Receiver A receiver for FPV video on the ground.

FPV Goggles Head-mounted displays for FPV flying.

FPV Monitor A monitor for FPV viewing.

Diversity Receiver A receiver using multiple antennas for better reception.

Antenna Polarization The orientation of electromagnetic wave propagation.

Linear Polarization Antenna polarization in a single plane.

Circular Polarization Antenna polarization rotating with the wave.

RHCP Right-hand circular polarization.

LHCP Left-hand circular polarization.

SMA Connector A threaded RF connector.

RP-SMA Reverse polarity SMA connector.

MMCX Connector A push-on RF connector.

UFL Connector A small surface-mount RF connector.

IPX Connector A miniature RF connector.

LQ Link quality metric.

SNR Signal-to-noise ratio.

Latency The delay between input and action.

End to End Latency Total delay from camera to goggles.

Frame Rate Frames per second of video.

Resolution Pixel dimensions of video, like 720p or 1080p.

Bitrate Data rate of video transmission.

Codec Video compression algorithm.

H.264 A widely used video codec.

H.265 A more efficient video codec.

Mavlink Protocol A protocol for communicating with drones.

Mavlink 2 The second version of the MAVLink protocol.

Mavlink Inspector A tool for inspecting MAVLink messages.

Parameter A configurable value on the autopilot.

Mission A sequence of waypoints and actions.

Flight Mode A configuration of the autopilot's control behavior.

Arming Enabling motor power on the drone.

Disarming Disabling motor power.

Pre-Arm Check Safety checks before arming.

Calibration Setting sensor offsets and scales.

Level Calibration Calibrating the accelerometer level.

Compass Calibration Calibrating the magnetometer.

ESC Calibration Calibrating the electronic speed controllers.

Radio Calibration Setting transmitter channel endpoints.

Throttle Calibration Calibrating the throttle channel.

Safety Switch A switch that must be activated before arming.

Beacon A sound or light for locating the drone.

Return to Launch A failsafe action returning to the launch point.

Land Immediately A failsafe action landing immediately.

Geofence Action The drone's response to geofence violation.

Battery Failsafe Action on low battery.

RC Failsafe Action on RC link loss.

GPS Failsafe Action on GPS signal loss.

Compass Error Action on compass failure.

Gyro Calibration Calibrating the gyroscope.

Accel Calibration Calibrating the accelerometer.

Space Science

Orbital mechanics, spacecraft engineering, and the space environment. From LEO to Lagrange points, re-entry to radiation belts.

Space Science The scientific study of the space environment, celestial bodies, and the universe.

Astronomy The study of celestial objects, space, and the universe as a whole.

Astrophysics The branch of astronomy dealing with the physics of celestial objects.

Cosmology The study of the origin, evolution, and structure of the universe.

Planetary Science The study of planets, moons, and planetary systems.

Exoplanet A planet orbiting a star outside our solar system.

Solar System The Sun and all objects gravitationally bound to it.

Star A luminous sphere of plasma held together by gravity.

Planet A large celestial body orbiting a star.

Moon A natural satellite orbiting a planet.

Asteroid A small rocky body orbiting the Sun.

Comet An icy body releasing gas and dust when near the Sun.

Meteor A streak of light from a meteoroid burning in the atmosphere.

Meteorite A meteoroid that reaches the Earth's surface.

Meteoroid A small rocky or metallic body in space.

Asteroid Belt A region between Mars and Jupiter containing many asteroids.

Kuiper Belt A region beyond Neptune with icy bodies and dwarf planets.

Oort Cloud A distant spherical shell of icy bodies surrounding the solar system.

Dwarf Planet A celestial body in hydrostatic equilibrium orbiting the Sun.

Pluto A dwarf planet in the Kuiper Belt.

Ceres The largest asteroid and a dwarf planet in the asteroid belt.

Sun The star at the center of our solar system.

Solar Flare A sudden eruption of energy on the Sun's surface.

Coronal Mass Ejection A massive burst of solar wind and magnetic fields.

Solar Wind A stream of charged particles released from the Sun.

Heliosphere The region of space dominated by the solar wind.

Magnetosphere The region around a planet influenced by its magnetic field.

Van Allen Belt A zone of energetic charged particles trapped by Earth's magnetic field.

Aurora A natural light display from solar wind interactions with the atmosphere.

Northern Lights Aurora Borealis in the northern hemisphere.

Southern Lights Aurora Australis in the southern hemisphere.

Ionosphere The ionized layer of the upper atmosphere.

Atmosphere The layer of gases surrounding a planet.

Exosphere The outermost layer of the atmosphere.

Thermosphere The atmosphere layer above the mesosphere with rising temperature.

Mesosphere The atmosphere layer above the stratosphere.

Stratosphere The atmosphere layer above the troposphere containing the ozone layer.

Troposphere The lowest atmosphere layer where weather occurs.

Ozone Layer A stratospheric layer absorbing UV radiation.

Greenhouse Effect Trapping of heat by atmospheric gases.

Carbon Dioxide A greenhouse gas important for planetary atmospheres.

Orbital Mechanics The study of the motion of objects in space.

Celestial Mechanics The branch of astronomy dealing with orbital motions.

Kepler's Laws The three laws of planetary motion.

Kepler's First Law Planets orbit in ellipses with the Sun at one focus.

Kepler's Second Law A line joining a planet to the Sun sweeps equal areas in equal times.

Kepler's Third Law The square of orbital period is proportional to the cube of semi-major axis.

Newton's Law of Gravitation Every mass attracts every other with force proportional to product of masses.

Gravitational Constant The constant G in Newton's gravitation law.

Gravity The force of attraction between masses.

Microgravity Near-weightlessness in free fall.

Weightlessness The experience of zero apparent weight.

Free Fall Motion under gravity alone.

Escape Velocity The speed needed to escape a gravitational field.

Orbital Velocity The speed required to maintain orbit.

Circular Velocity The speed for a circular orbit.

Hohmann Transfer Orbit A fuel-efficient transfer between two circular orbits.

Bi-Elliptic Transfer A transfer using two elliptical arcs.

Gravity Assist Using a planet's gravity to change velocity.

Swing-By Another term for gravity assist.

Slingshot Informal term for a gravity assist trajectory.

Lagrange Point A point where gravitational forces balance in a two-body system.

L1 The first Lagrange point between Sun and Earth.

L2 The second Lagrange point beyond Earth from the Sun.

Lissajous Orbit A quasi-periodic orbit around a Lagrange point.

Halo Orbit A periodic orbit around a Lagrange point.

Patched Conics Approximating interplanetary trajectories as conic sections.

Sphere of Influence The region where one body's gravity dominates.

Launch Window A favorable period for launching to a target.

Interplanetary Transfer A trajectory between planets.

Earth Orbit An orbit around the Earth.

Low Earth Orbit An orbit below 2,000 km altitude.

Medium Earth Orbit An orbit between 2,000 and 35,786 km.

Geosynchronous Orbit An orbit with a 24-hour period.

Geostationary Orbit A circular equatorial geosynchronous orbit at 35,786 km.

Geostationary Transfer Orbit An elliptical orbit used to reach GEO.

Highly Elliptical Orbit An elongated orbit with a high apoapsis.

Molniya Orbit A highly elliptical orbit with 12-hour period.

Tundra Orbit A geosynchronous highly elliptical orbit.

Polar Orbit An orbit passing over the poles.

Sun-Synchronous Orbit A polar orbit with constant Sun angle.

Dawn-Dusk Orbit A Sun-synchronous orbit at the terminator.

Frozen Orbit An orbit with minimal perturbation effects.

Graveyard Orbit A disposal orbit for decommissioned satellites.

Parking Orbit A temporary orbit before further maneuvers.

Phasing Orbit An orbit adjusting the relative position.

Co-orbital Sharing the same orbit.

Retrograde Orbit An orbit opposite the planet's rotation.

Prograde Orbit An orbit in the direction of the planet's rotation.

Ascending Node The point where an orbit crosses the reference plane northward.

Descending Node The point where an orbit crosses the reference plane southward.

Line of Nodes The line connecting ascending and descending nodes.

Argument of Perigee The angle from the ascending node to perigee.

Right Ascension of Ascending Node The angle defining the orbital plane orientation.

True Anomaly The angular position along the orbit from periapsis.

Mean Anomaly The angular position from periapsis for uniform circular motion.

Eccentric Anomaly The angular parameter relating true and mean anomaly.

Semi-Major Axis Half the longest diameter of an elliptical orbit.

Semi-Minor Axis Half the shortest diameter of an elliptical orbit.

Eccentricity The deviation of an orbit from circular.

Inclination The tilt of an orbit relative to the reference plane.

Periapsis The closest point in an orbit to the central body.

Apoapsis The farthest point in an orbit from the central body.

Perigee Periapsis for Earth orbits.

Apogee Apoapsis for Earth orbits.

Perihelion Periapsis for solar orbits.

Aphelion Apoapsis for solar orbits.

Orbital Period Time for one complete orbit.

Synodic Period The time between two alignments of a body with the Sun.

Sidereal Period The orbital period relative to the fixed stars.

Nodal Precession The rotation of the orbital plane due to perturbations.

Apsidal Precession The rotation of the orbit's major axis.

Perturbation A deviation from ideal orbital motion.

J2 Perturbation The perturbation from a planet's equatorial bulge.

Third Body Perturbation Gravitational perturbation from another body.

Drag Perturbation Atmospheric drag reducing orbit altitude.

Radiation Pressure Force from solar radiation on a spacecraft.

Poynting-Robertson Effect Drag from solar radiation on dust particles.

Yarkovsky Effect A force on rotating asteroids from thermal emission.

Roche Limit The distance within which a moon will be torn apart by tidal forces.

Hill Sphere The region where a body's gravity dominates over the Sun.

Tidal Force A force stretching a body in a gravitational gradient.

Tidal Locking A body with the same face always toward its primary.

Synchronous Rotation Rotation matching the orbital period exactly.

Spacecraft A vehicle designed for operation in space.

Satellite A man-made object placed in orbit.

Space Station A habitable artificial satellite in orbit.

ISS International Space Station.

Capsule A crewed or cargo spacecraft with a blunt body shape.

Space Probe An unmanned spacecraft for exploration.

Lander A spacecraft that lands on a celestial body.

Rover A mobile exploration vehicle on a celestial body.

Orbiter A spacecraft that orbits a celestial body.

Flyby Spacecraft A spacecraft passing near a target without orbiting.

Sample Return Mission A mission returning samples to Earth.

Crewed Spacecraft A spacecraft carrying humans.

Space Suit A garment protecting astronauts in space.

EVA Suit An Extravehicular Activity suit for spacewalks.

Life Support System A system providing air, water, and temperature control.

ECLSS Environmental Control and Life Support System.

Habitat Module A module providing living quarters in space.

Airlock A chamber for passing between different environments.

Docking Mechanism A system connecting spacecraft together.

Androgynous Docking A docking system where both sides are identical.

Probe and Drogue A docking system with a probe and receiving cone.

Capture The moment of first contact during docking.

Soft Capture Initial docking with minimal force.

Hard Capture Final latched docking connection.

Berthing Moving a module to a docking port using a robotic arm.

Robotic Arm A manipulator arm on a spacecraft or station.

Canadarm The robotic arm on the ISS.

Remote Manipulator System A robotic arm for satellite deployment and retrieval.

Space Tether A long cable connecting spacecraft in orbit.

Electrodynamic Tether A conductive tether generating current through the magnetic field.

Solar Sail A spacecraft using sunlight pressure for propulsion.

Solar Panel A device converting sunlight to electricity.

Photovoltaic Cell A cell directly converting light to electricity.

Solar Array An assembly of solar panels on a spacecraft.

RTG Radioisotope Thermoelectric Generator.

Radioisotope Thermoelectric Generator A generator using heat from radioactive decay.

Nuclear Power in Space Using nuclear reactors or RTGs for space power.

Battery An electrochemical energy storage device.

Fuel Cell A device generating electricity from fuel and oxidizer.

Attitude Control System A system controlling spacecraft orientation.

Reaction Wheel A flywheel for attitude control.

Control Moment Gyro A gyroscope for attitude control.

Thruster A small rocket engine for attitude control.

Reaction Control System Thrusters for spacecraft attitude and translation.

Star Tracker An optical sensor for star-based attitude determination.

Sun Sensor A sensor determining the direction to the Sun.

Horizon Sensor A sensor detecting the planet's horizon.

Earth Sensor A sensor detecting Earth for attitude reference.

Magnetometer A sensor measuring magnetic field in space.

Gyroscope A sensor measuring angular rate.

Inertial Measurement Unit A unit combining accelerometers and gyroscopes.

Attitude Determination Calculating spacecraft orientation from sensor data.

Attitude Propagation Predicting orientation over time from known state.

Despin Reducing spin rate of a spacecraft.

Spin Stabilization Stabilizing a spacecraft by spinning.

Three-Axis Stabilization Active stabilization in all three axes.

Gravity Gradient Stabilization Using gravity differences for stabilization.

Magnetic Torquer A device using magnetic fields for attitude control.

Magnetorquer An electromagnetic coil for attitude control.

Thermal Control System A system maintaining spacecraft temperature.

Passive Thermal Control Using insulation, coatings, and radiators.

Active Thermal Control Using heaters and pumped fluid loops.

Multi-Layer Insulation Reflective insulation blankets for spacecraft.

Radiator A surface radiating heat to space.

Heat Pipe A passive heat transfer device.

Louver A variable-position radiator cover.

Phase Change Material A material absorbing heat through melting.

Thermal Blanket An insulating cover for spacecraft.

Cold Plate A plate cooling electronics.

Telemetry Data transmitted from the spacecraft to ground.

Telecommand Commands transmitted from ground to the spacecraft.

Ground Station A facility communicating with spacecraft.

Antenna A communication antenna on the spacecraft.

High Gain Antenna A directional antenna for high data rates.

Low Gain Antenna An omnidirectional antenna.

Parabolic Antenna A dish antenna for focused communication.

Phased Array An electronically steered antenna.

Transponder A device receiving and retransmitting signals.

RF Communication Radio frequency communication with spacecraft.

S Band A frequency band around 2-4 GHz for spacecraft.

X Band A frequency band around 8-12 GHz for high data rate.

Ka Band A frequency band around 26-40 GHz for very high data rates.

Deep Space Network NASA's network for deep space communication.

DSN Deep Space Network.

Data Rate The rate of data transmission from spacecraft.

Compression Reducing data size for transmission.

Error Correction Coding for reliable data transmission.

Reed-Solomon Code An error correction code for space communications.

Convolutional Code An error correction code using convolution.

Turbo Code A high-performance error correction code.

LDPC Low-Density Parity-Check code for near-capacity performance.

Spacecraft Bus The primary structure carrying all spacecraft subsystems.

Structure The mechanical framework of a spacecraft.

Honeycomb Panel A lightweight panel for spacecraft structures.

Aluminum Honeycomb A core material for spacecraft panels.

Carbon Fiber Structure A lightweight composite spacecraft structure.

Deployable Structure A structure unfolding in space.

Solar Array Deployment Unfolding of the solar panels.

Antenna Deployment Unfolding of the communication antenna.

Boom A deployable arm or mast on a spacecraft.

Mast A deployable vertical structure on a spacecraft.

Mechanism A moving part on a spacecraft.

Pyrotechnic Device An explosive device for separation or deployment.

Frangible Nut A pyrotechnic nut for releasing mechanisms.

Cable Cutter A pyrotechnic cutter for release cables.

Pin Puller A pyrotechnic device retracting a pin.

Separation Nut A nut released pyrotechnically for stage separation.

Launch Adapter An interface between spacecraft and launch vehicle.

Payload Adapter A structure attaching the payload to the stage.

Clamp Band A band holding the payload until separation.

Marman Clamp A clamp band system for spacecraft separation.

Lightband A low-shock separation system.

Separation System A system releasing the spacecraft from the launch vehicle.

Yo-Yo Deploy A technique for reducing spin before separation.

Spin Table A rotating platform for spinning up payloads.

Balance Mass balancing for stability.

Center of Mass The balance point of the spacecraft.

Margin of Safety A factor of safety in structural design.

Qualification Model A test model for design validation.

Flight Model The actual spacecraft to be launched.

Engineering Model A test model for engineering development.

Structural Model A model for structural testing.

Thermal Model A model for thermal testing.

Protoflight Model A model combining qualification and flight.

Clean Room A controlled environment for spacecraft assembly.

Cleanliness Control of particulate contamination.

Outgassing Release of gas from materials in vacuum.

Vacuum The absence of matter in space.

Thermal Vacuum Test Testing in simulated space conditions.

Vibration Test Testing under launch vibration.

Acoustic Test Testing under launch acoustic levels.

Shock Test Testing under pyrotechnic shock.

EMI/EMC Test Testing electromagnetic compatibility.

Mass Properties Measurement Measuring mass, center of mass, and inertia.

Spin Balance Balancing a spinning spacecraft.

Alignment Optical alignment of spacecraft instruments.

Integration Assembling the spacecraft subsystems.

Test Verifying spacecraft functionality.

Launch Campaign Activities from arrival at launch site to liftoff.

Launch Site A facility for launching spacecraft.

Cape Canaveral A major US launch site in Florida.

Kennedy Space Center NASA's launch site in Florida.

Vandenberg SFB A US West Coast launch site.

Wallops Flight Facility A NASA launch site in Virginia.

Baikonur A Russian launch site in Kazakhstan.

Kourou A European launch site in French Guiana.

Spaceport A facility for launching spacecraft.

Launch Pad A platform for launching.

Launch Complex A facility with launch pad and support.

Integration Facility A building for spacecraft and rocket integration.

Payload Processing Preparing the payload for launch.

Propellant Loading Fuelling the spacecraft or stage.

Countdown The timeline before launch.

Big Bang The cosmological event that began the expansion of the universe.

Dark Matter An unseen form of matter inferred from its gravitational effects.

Dark Energy A mysterious force causing the accelerated expansion of the universe.

Black Hole An object with gravity so strong that nothing, not even light, can escape.

Neutron Star An extremely dense stellar remnant composed primarily of neutrons.

White Dwarf A stellar remnant supported by electron degeneracy pressure.

Supernova A catastrophic stellar explosion that can outshine an entire galaxy.

Nebula An interstellar cloud of dust, hydrogen, helium, and plasma.

Galaxy A massive system of stars, stellar remnants, gas, dust, and dark matter.

Milky Way The galaxy containing our Solar System.

Light-Year The distance light travels in one year, about 9.46 trillion km.

Astronomical Unit The mean distance from Earth to the Sun, about 150 million km.

Parsec A unit of distance equal to about 3.26 light-years.

Redshift The stretching of light to longer wavelengths due to cosmic expansion.

Cosmic Microwave Background The relic radiation from the Big Bang, now at microwave wavelengths.

Terrestrial Planet A rocky planet composed primarily of silicate rocks and metals.

Gas Giant A large planet composed mostly of hydrogen and helium.

Impact Crater A depression on a celestial body caused by a meteoroid impact.

Regolith The loose, unconsolidated surface material covering solid celestial bodies.

Cubesat A standardized small satellite format, typically 10 cm per unit.

SmallSat A category of small satellites with reduced mass and size.

Space Tourism Commercial spaceflights for recreational, leisure, or business purposes.

Planetary Protection Preventing biological contamination of celestial bodies during exploration.